

Classifying Suicide-Related Communication on Reddit

Can TF-IDF models accurately classify Reddit posts across five suicide-related categories, and which features are most informative?



Ruby Villalona, Libby Barak,
Department Of Linguistics & School of Computing • villalona.r@montclair.edu

Motivation

- Subtle linguistic signs of suicidal ideation are present in Reddit posts, but these signals are often difficult to detect manually.
- Moderators and crisis teams cannot review all content on large platforms like Reddit, causing high-risk posts to be overlooked.
- Identifying linguistic cues across different risk levels can improve early crisis detection, help prioritize high-risk posts, and create safer online environments.
- Existing NLP models may overweight ideation language while undervaluing real behavioral risk, leading to dangerous misclassifications.
- Bias-aware classification is necessary to avoid critical failures in crisis detection and to better support at-risk individuals.

Background

Prior work shows that suicide risk can be inferred from language in Reddit posts. The CLPsych 2019 shared task emphasized multi-level classification rather than simple binary labels.

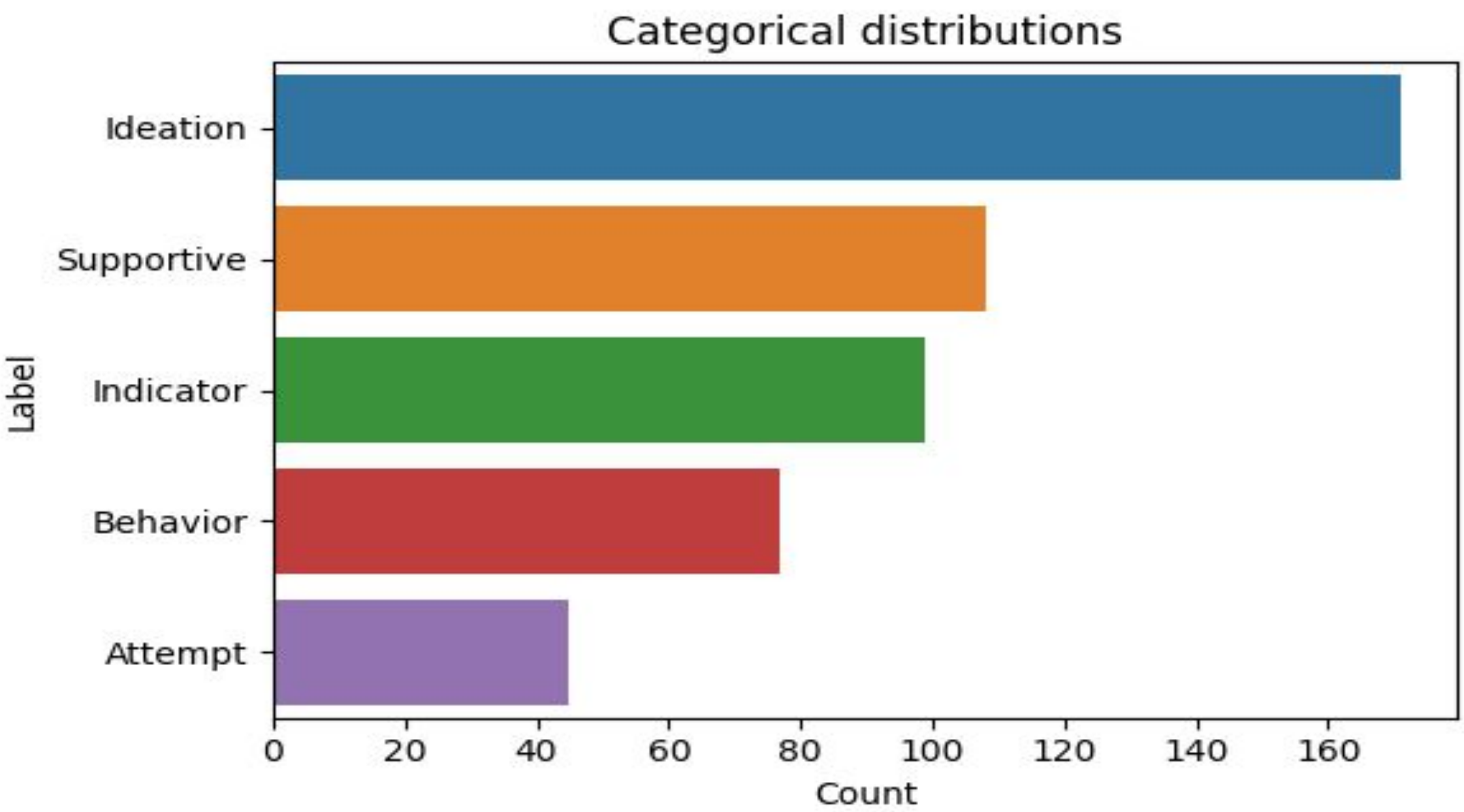
Studies indicate that Supportive and Ideation posts are easier to classify, while Behavior and Attempt categories are more difficult due to overlapping language. This overlap often leads to model confusion.

TF-IDF combined with linear models is commonly used as a strong and interpretable baseline.

TF-IDF (Term Frequency–Inverse Document Frequency) highlights words that are important to a text but not common overall

Data

- 500 anonymized Reddit mental-health posts (one per unique user)
- Labeled into five categories:
Supportive, Ideation, Behavior, Attempt, Indicator
- Text is short–medium length, informal, emotional, and conversational
- Contains real expressions of distress, coping, and suicide-related communication
- Supervised dataset with some class imbalance
- Minimal cleaning needed before analysis (raw text with Reddit slang + punctuation)
- Suitable for studying linguistic cues across different suicide-risk levels



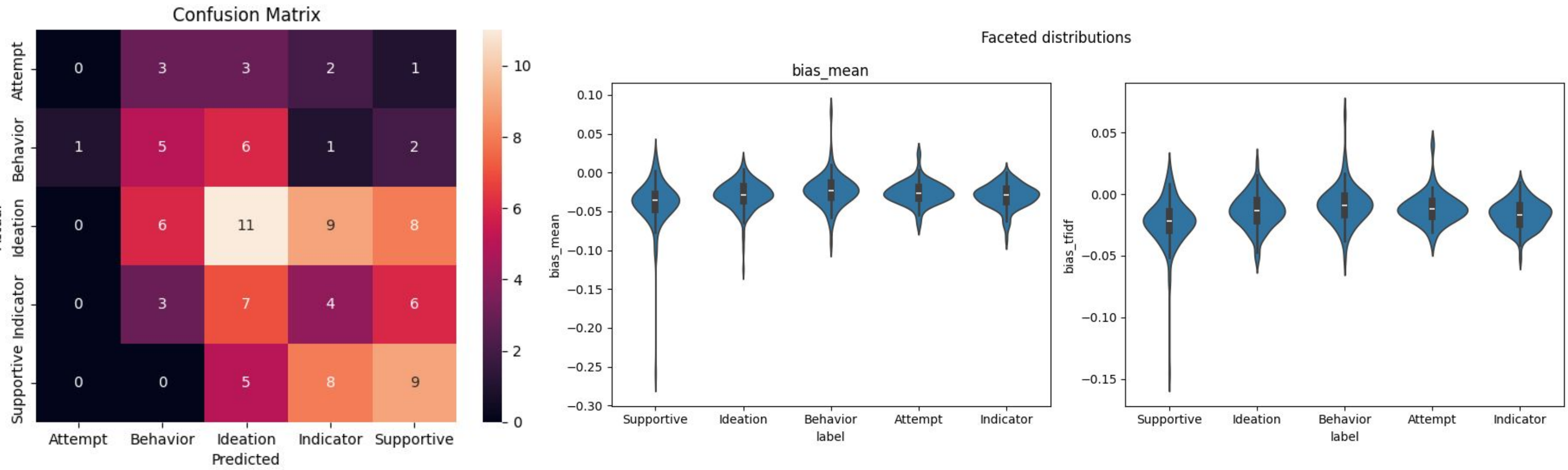
Method

- The project uses the provided TF-IDF-based text classification approach.
- Posts are converted into TF-IDF feature vectors
- A multiclass Logistic Regression model is trained to classify posts into:
Supportive, Ideation, Behavior, Attempt, Indicator.
- The dataset is preprocessed (cleaning, tokenizing, splitting into train/validation/test).
- Evaluation uses accuracy, precision, recall, F1, and confusion matrices.
 - Accuracy = overall correctness | Precision = correctness of positives | Recall = coverage of positives

Raw Text → Preprocess → Train/Test Split → TF-IDF → LR Model → Bias Analysis
Text classification pipeline

Attempt: attempt, somebody, suicide attempt, failed
Behavior: cutting, hospital, felt, kill im
Ideation: thank, depression, mental depression, mental
Indicator: behavior, hyperactive, hyperactive behavior, haha
Supportive: think, sounds, therapist help, chocolate

Results



- Supportive posts are the most linguistically distinct and best classified (9 correct), while Attempt posts are never correctly classified (0), showing the model fails on the highest-risk class
- Ideation and Indicator are the most confused categories, frequently misclassified across multiple classes due to overlapping language
- TF-IDF bias is more compressed than mean bias, showing it partially reduces but does not eliminate bias across all classes
- Attempt posts show the tightest bias distribution, while Supportive posts show the widest, indicating immediate-risk language is more emotionally consistent

Conclusion

- ❑ **Language alone can reveal different suicide-risk levels** in Reddit posts.
- ❑ **TF-IDF is a strong, interpretable baseline** for mental-health text classification.
- ❑ **Supportive and Ideation posts are easier to identify** than Behavior and Attempt.
- ❑ **Class imbalance and language ambiguity matter** for model performance.
- ❑ Understanding these patterns can **support safer online spaces and early intervention**.

References

Zirikly, A., Resnik, P., Uzuner, Ö., & Hollingshead, K. (2019). CLPsych 2019 Shared Task: Predicting the Degree of Suicide Risk in Reddit Posts. Proceedings of the Sixth Workshop on Computational Linguistics and Clinical Psychology.